

# Crypto Market Sentiment & Trader Performance Analysis

## 1. Objective

Analyze the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader performance using Hyperliquid trading data.

## 2. Datasets

### Fear & Greed Index Dataset

- Date
- Classification (Fear, Greed, Extreme Fear, Extreme Greed, Neutral)

### Hyperliquid Historical Trading Dataset

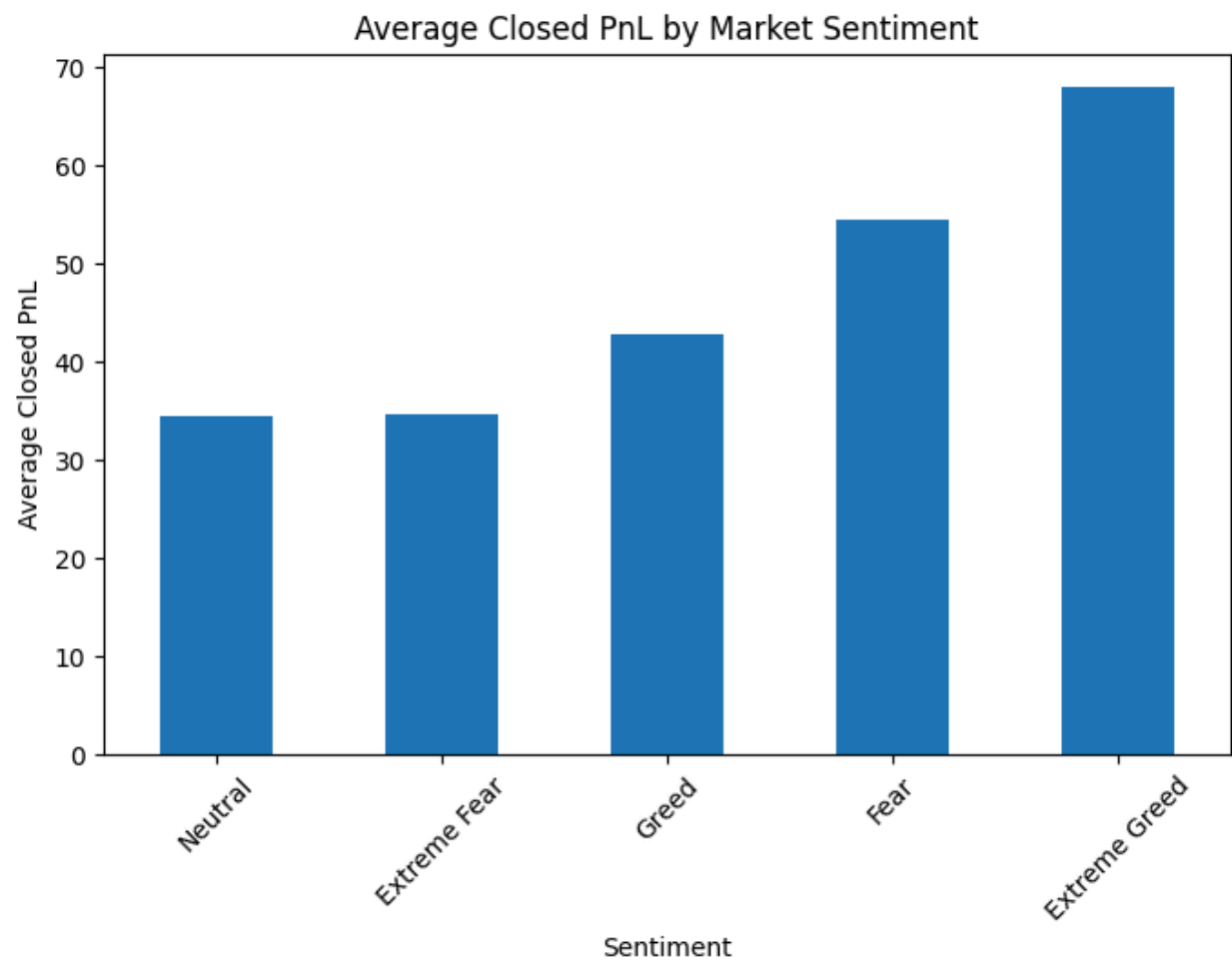
- Account
- Coin
- Execution Price
- Size USD
- Closed PnL
- Fee
- Timestamp

## 3. Data Processing

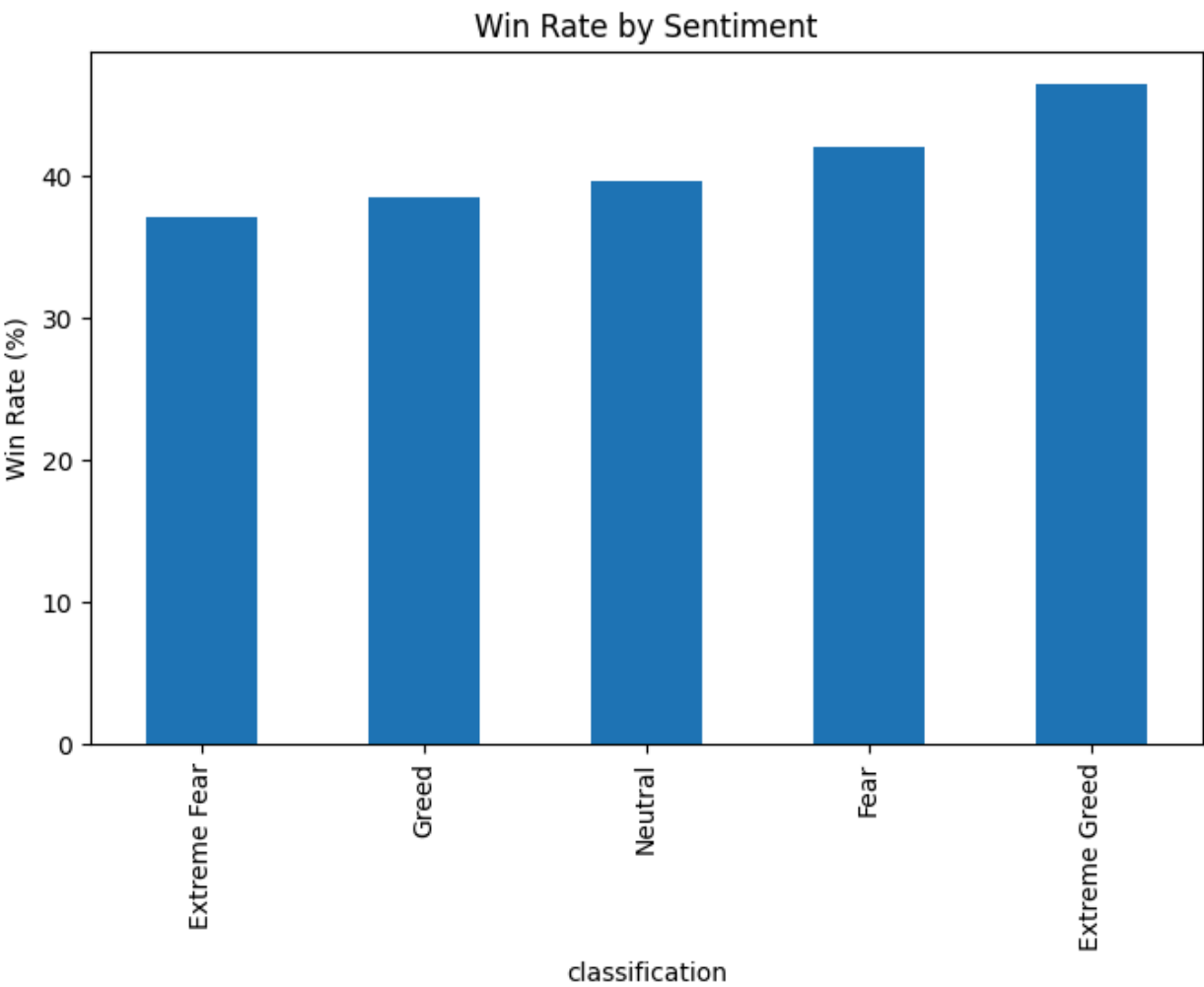
- Cleaned datasets
- Converted timestamps to dates
- Merged datasets using common dates
- Total merged records: **211,218**
- Unique traders: **32**
- Unique coins: **246**

## 4. Key Visualizations

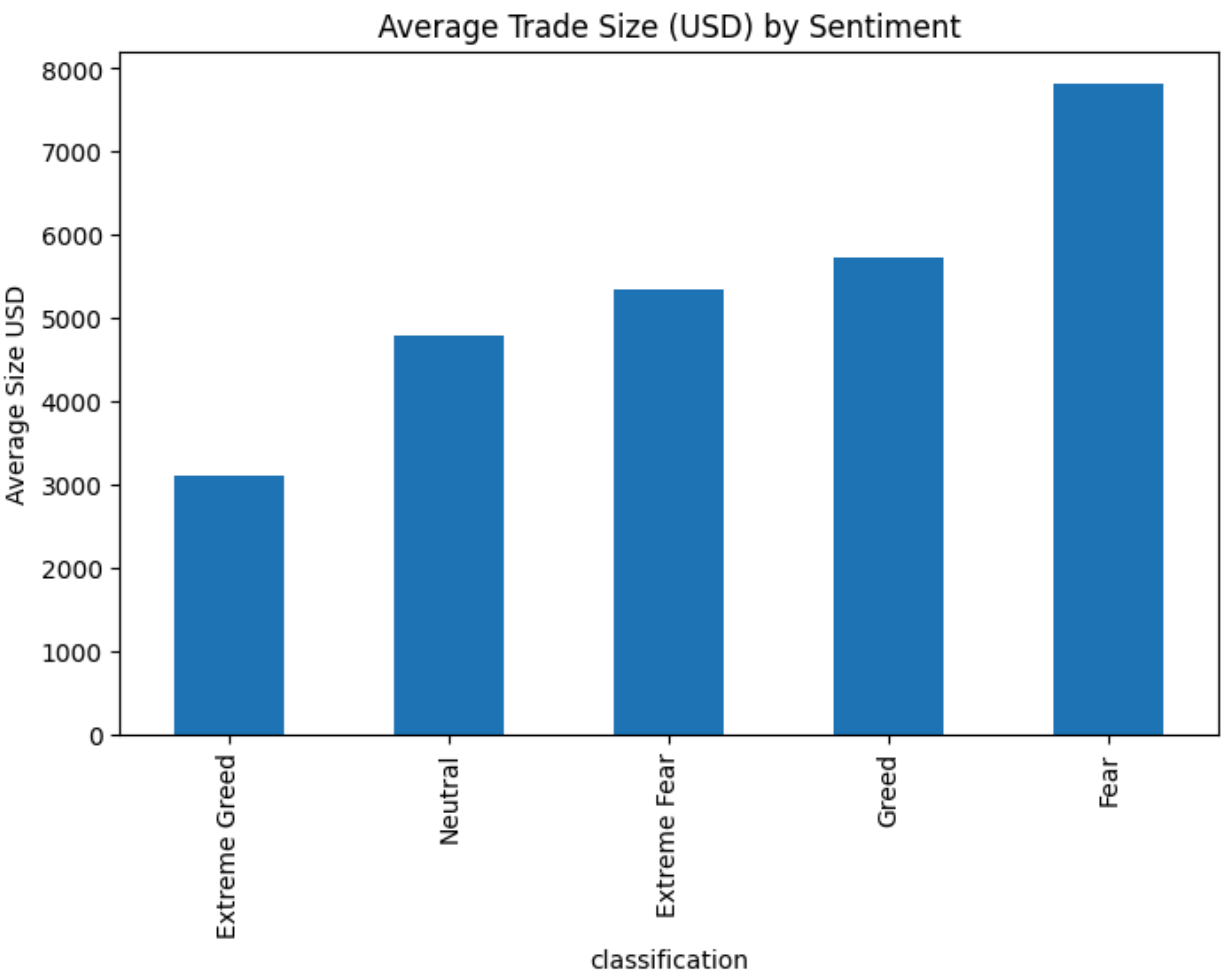
### Average Closed PnL by Sentiment



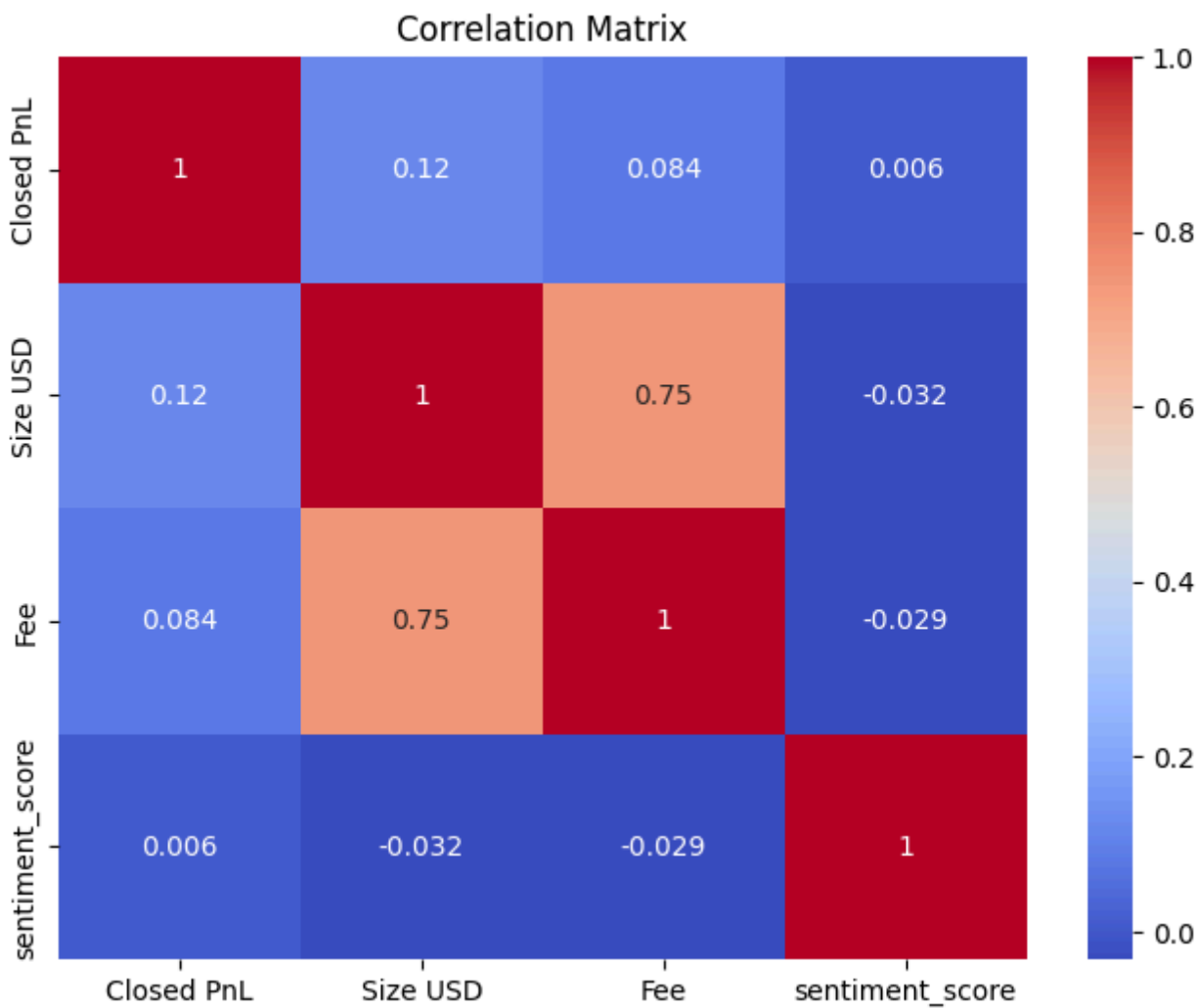
Win Rate by Sentiment



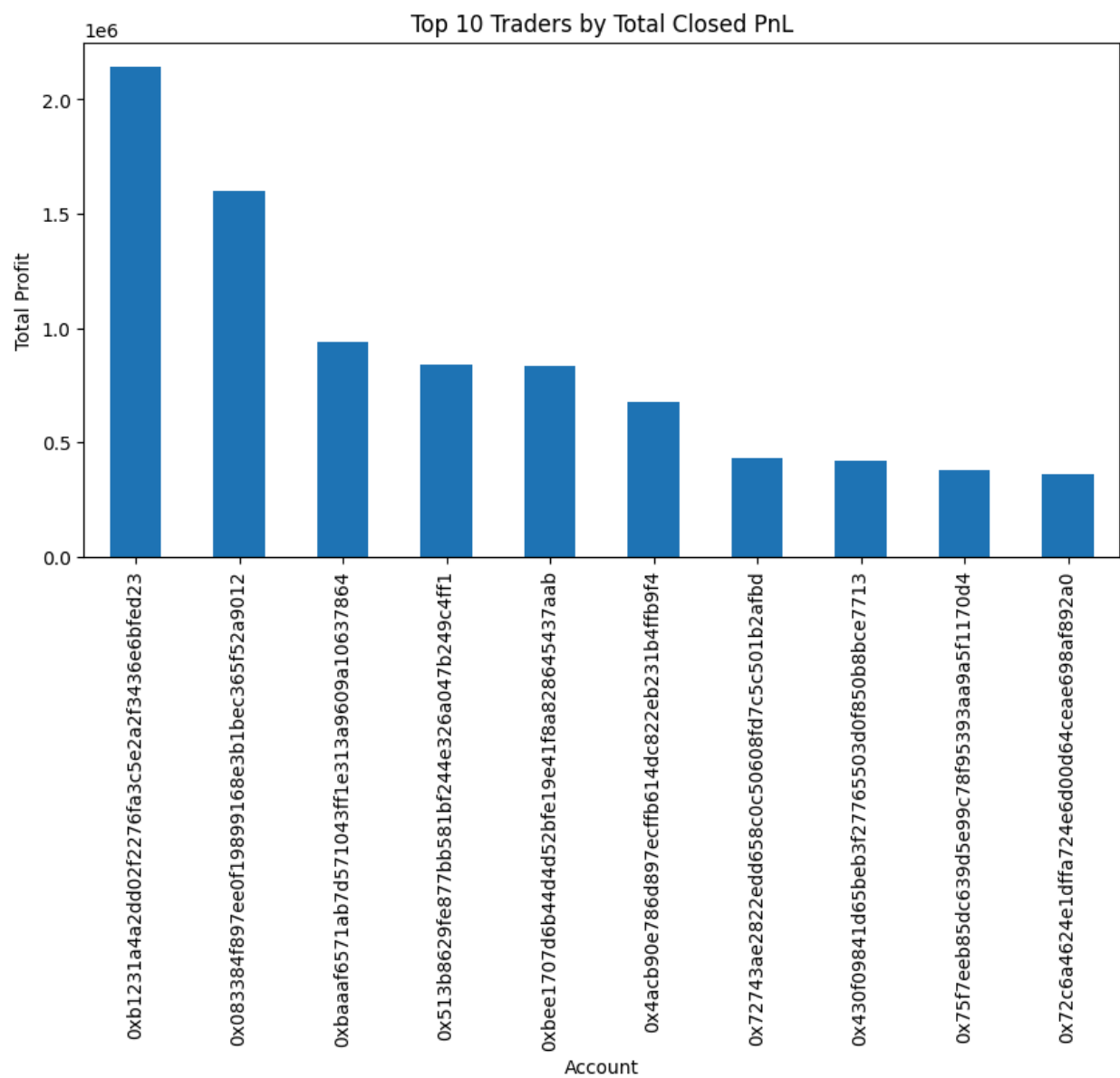
Average Trade Size by Sentiment



Correlation Heatmap



Top Traders by Closed PnL



## 5. Statistical Testing

### ANOVA Test Results

- F-statistic: **7.3176**
- P-value: **0.000067**
- Since  $p\text{-value} < 0.05$ , trader profitability differs significantly across sentiment categories.

## 6. Key Findings

1. Extreme Greed showed the highest average profitability.
2. Win rates increased during optimistic market conditions.
3. Fear periods showed the largest average trade sizes.
4. Sentiment had a statistically significant impact on profitability.
5. Profitability was concentrated among a small group of traders.

## 7. Conclusion

Market sentiment influences trader behavior and profitability. While sentiment is not a strong linear predictor of profit, significant differences exist across sentiment regimes. Traders can use sentiment as an additional signal for risk management and decision-making.